MIL-C-81185B(AS)
1 March 1967
Superseding
MIL-C-81185A(WP)
30 April 1966

MILITARY SPECIFICATION

Computer Set, Navigational

AN/ASN-41(*)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall compute and display the present position of an aircraft in flight and shall compute the course and distance to a pre-selected destination.

1.2 Classification - The Navigation Computer Set shall consist of the items listed in 6.9.

2. APPLICABLE DOCUMENTS

* 2.1 General - The documents listed below of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases; Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

FSC 5826

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-23603	System Readiness/Maintainability; Avionic Systems Design, General Specification for

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MS17322	Meter, Time Totalizing, 115 Volt, 400-Cycle

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2.1.1 Availability of Documents -

(1) When requested specifications, standards, drawings and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

* 3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

*3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of AR-5.

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*3.2.3 Microelectronic Interchangeability - The interchangeability provisions of AR-5 shall govern Microelectronic interchangeability.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 36 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 3000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 200 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

Unit	Type of Meter
C-6243/ASN-41	MS-17322
CP-774/ASN-41	MS-17322

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

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3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability of Reordered Equipment - For re-ordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115±1.0 V AC

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

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3.3.10.1 Vibration - The Computer, Navigational CP-774/ASN-41 shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400. The Control C-6243/ASN-41 shall operate satisfactorily when subjected to the vibration requirements of Curve III of Specification MIL-E-5400.

3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 2 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Primary Inputer Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

AC Power (Single Phase) 115V, Category "B", 175 VA

3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all circuits except the integrators.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The equipment shall accept electrical inputs of the ground speed, drift angle, magnetic heading and true air speed; manual inputs of present position, destination position, and magnetic variation; shall continuously compute and display the present position in latitude and longitude and shall compute the distance and relative bearing from the present position to the destination. The equipment shall also provide an electrical output of true heading. The computation of distance and course to destination shall be a planar solution for distances between zero and 195 nautical miles and a great circle solution for distances between 195 and 1999 nautical miles. When doppler derived ground velocity is available, the equipment shall compute and display wind speed and wind "from" direction.

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3.4.2 Input Signals - The equipment shall accept electrical and manual input signals as shown in Tables 1 and 2.

3.4.3 Outputs - The equipment shall provide electrical and visual outputs as listed in Tables 3 and 4. The electrical output signals shall be compatible with the input requirements of the ID-663 ( )/U Bearing-Distance-Heading Indicator. The electrical output signals shall be capable of operating either one ID-663/U or two ID-663/U Indicators connected in parallel.

3.4.4 Accuracy - The accuracy of the computations performed by the equipment shall be as shown in Tables 3 and 4. The stated accuracies shall be maintained with no adjustments and shall be valid for all service conditions specified in 3.3.10.

3.4.4.1 Earth Oblateness Correction - Earth oblateness correction shall be mechanized within the computer to provide solution accuracies within the specified tolerances.

3.4.5 Modes of Operation - The equipment shall have the following modes of operation.

3.4.5.1 Doppler Mode - In the Doppler Mode the equipment shall accept inputs of ground speed, drift angle, true airspeed, magnetic heading and magnetic variation to form the true ground velocity vector and true air velocity vector. The true ground velocity vector shall be used for dead reckoning and in addition shall be added to the true air velocity vector to provide a continuous solution of the wind vector. All system outputs shall be available in this mode.

3.4.5.2 Memory Mode - The memory mode of operation shall be automatically actuated when the memory signal is applied indicating loss of doppler operation. The Wind Memory portion of the Computer Set shall retain the last value of the wind vector computed prior to the loss of doppler operation. The computed Wind Vector shall be added to the true air velocity vector to form a computed true ground velocity vector to be used for dead reckoning computations. All outputs, except the computed wind direction and wind speed, shall be available in this mode of operation.

3.4.5.3 Air Mass Mode - In the Air Mass Mode the Computer Set shall receive inputs of true air speed, magnetic heading and magnetic variation. The wind speed and wind direction shall be inserted manually. The true air speed, the heading and the wind vector shall be added to form a true ground velocity vector to be used for the dead reckoning. All outputs, except the computed wind direction and wind speed, sh ll be available in this mode of operation.

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TABLE 1

Electrical Input Signal Characteristics

Input	Range	Synchro type (from)	Scale Factor	Zero	Maximum Rate of Change
Ground Speed	50 to 990 kts	Synchro Kearfott RS910-1 or equivalent	36°/100 kts	Synchro zero at 550 kts	160 kts/sec
Drift Angle	40° right to 40° left	Synchro Kearfott RS910-1 or equivalent	1°/1°	Synchro zero at 0°	20°/sec
True Air Speed	50 to 800 kts	Synchro Bendix AY-503-38 or equivalent (Note 1)	36°/100 kts	Synchro zero at 50 knots	160 kts/sec
Magnetic Heading	0° to 360°	Synchro Bendix AY-500-56 or equivalent	1°/1°	Synchro zero at 0°	20°/sec
Doppler Memory	On or off	DC voltage	28V ± 1Vdc	See Note 3	

Note 1 - When viewed from the shaft end, CCW rotation denotes increasing air speed.

Note 2 - All synchros are excited by the Navigation Computer Set.

Note 3 - The 28V DC Doppler Memory signal is "on" when the doppler is on.

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TABLE 2

Manual Input Characteristics

Input	Characteristic	
Present Position	Latitude	80° 00'S to 80° 00'N
	Longitude	180° 00'E to 180° 00'W
Destination	Latitude	80° 00'S to 80° 00'N
	Longitude	180° 00'E to 180° 00'W
Magnetic Variation	180°E to 180°W	
Wind Speed	0 to 200 kts	
Wind Direction	0 to 360°	

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*TABLE 3

Electrical Output Signal Characteristics

Output	Range	Scale Factor	Form	Accuracy
Distance to Destination	0 to 1999NM	Flag Signal for 1000NM	Close Contacts to ground	0-195 NM: 1.5 NM or 1.5% whichever is greater 195-1999 NM: 7.5 NM +1.5%
		Three synchros, one for each decimal digit Units 0 to 9NM Tens 0 to 90NM Hundreds 0 to 900NM	26V, 400cps	
Relative Bearing to Destination	0° to 360°	1°/1° Synchro	26V, 400cps	±1.5° or consistent with termination error whichever is greater
Relative Ground Track	45°L to 45°R	1°/1° Synchro	26V, 400cps	±1.5°
True Heading	0° to 360°	1°/1° Synchro	26V, 400cps	±0.375

Note 1 - The excitation for the output synchros shall be provided by the using equipment.

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TABLE 4

Visual Output Characteristics

Output	Range		Form	Accuracy
Wind Speed	0 to 200 kts		Counter	1 knot or 0.3% of ground speed
Wind Direction	0° to 360°		Counter	±1.0° or consistent with wind speed tolerance whichever is greater
Present Position	Latitude Longitude	80°N to 80°S 180°E to 180°W	Counters	Latitude or Longitude 1 NM per hour or P times distance traveled whichever is greater Where: P= ± 0.25% in latitude P= ± 0.25% for longitude computation between 60°N and 60°S latitude P= ± 0.5% for longitude computation between 60°N&S and 70° N&S latitude P= ± 1.0% for longitude computation between 70° N&S and 80° N&S latitude

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3.4.6 Position Changes - It shall be possible to change the destination counters while the Computer Set is in operation. Changing the setting of the destination counters shall not interrupt or distrub the computation of present position.

3.4.7 Self Test - The equipment shall include a front panel mounted test switch permitting integrity testing of the Computer Set without requiring power application to associated subsystems. A standard problem shall be applied to the system to enable verification of operation of the wind computer as well as the present position computation, by monitoring the various computer-control panel counters. Verification of the computation of range and relative bearing shall be by monitoring the ID-663/U Indicator.

3.4.8 Internal Scale Factors - All mechanical shaft position scale factors and electrical scale factors shall be constant from one equipment to another and from one sub-assembly to another such that no adjustment is required to maintain complete interchangeability of identical modules within the units of the Navigation Computer Set, interchangeability of units within the Navigation Computer Set and interchangeability of Navigation Computer Sets.

3.4.9 Module Interchangeability - All sub-assemblies shall be module constructed and shall be capable of being serviced independently and shall be completely interchangeable between equipments with no adjustments required to meet the performance requirements of this specification.

3.4.9.1 Module Serviceability - Each module shall be easily and independently removable for maintenance. The number of fasteners having to be removed to remove each module shall not exceed four. The removal of any one module from a unit shall not require the removal of any other module or the main wiring harness.

3.4.10 Control Panel Lighting - The Control Panel Lighting shall be red light in accordance with MIL-P-7788. The Control Panel Lighting shall operate from 28V AC electric power. All bulbs shall be replaceable.

3.4.11 Wind Averaging - Computed wind speed and direction shall be displayed on the Control CP-774/ASN-41. The displayed wind shall be a continuous average of the instantaneous wind. Wind averaging shall be accomplished by limiting the maximum rate of change of wind speed to 2.4 knots per second.

3.4.12 Part Interchangeability - All parts having identical part numbers, or parts with different part numbers performing identical functions and having identical circuit designations shall be constructed to permit complete interchangeability in order to preclude the modification, replacement, or adjustment of related circuit components when being serviced. This does not include the possible remulling of synchros, etc., in common gear trains with the serviced part where nulls of such

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other components may be disturbed. In no case shall the replacement of a component in one sub-assembly require the modification, replacement, or adjustment of another component in another sub-assembly.

3.5 Detail Requirements -

3.5.1 Control-Indicator C-6243/ASN-41 - The Control-Indicator shall meet the applicable requirements of Specification MIL-C-6781 for Type I control panels and, in addition, the following requirements:

3.5.1.1 Function - The Control-Indicator shall be the unit into which the navigation problem is manually inserted. It shall display the Present Position of the carrying aircraft in Latitude and Longitude, the Destination Coordinates in Latitude and Longitude, the Wind Direction in degrees, the Wind Velocity in Knots, and the Variation in degrees. Controls shall be provided for setting these indications. A Master Selector Switch shall be provided to control the operation of the Navigation Computer Set. The unit shall operate with the Computer, Navigational utilizing manual inputs and automatic inputs to the system to provide the over-all requirements.

3.5.1.2 Form Factor - The Control-Indicator shall be console mounted and shall conform to the dimensions shown in Figure 2.

3.5.1.3 Weight - The weight of the Control-Indicator shall not exceed 10 pounds.

3.5.1.4 Displays - The Control-Indicator shall contain the following displays:

3.5.1.4.1 Position Latitude and Longitude - Present position latitude and longitude shall be displayed on counters. The counters shall continually change to display the present position of the aircraft during flight. The display shall be in degrees and minutes of latitude and longitude, in one minute increments.

3.5.1.4.2 Destination Latitude and Longitude - Destination coordinates in latitude and longitude shall be displayed on counters. The display shall be in degrees and minutes of latitude and longitude in one minute increments. The counters shall display the coordinates of either Destination 1 or Destination 2 as selected by the Master Selector Switch.

3.5.1.4.3 Magnetic Variation - Magnetic Variation shall be displayed on counters in degrees and tenths of degrees in one tenth of a degree increments.

3.5.1.4.4 Wind Direction - The wind direction counters shall display the true wind "from" direction. The display shall be in degrees in one degree increments.

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3.5.1.4.5 Wind Speed - The magnitude of wind speed shall be displayed on counters. The display shall be in knots in one knot increments.

3.5.1.5 Controls - The control-indicator shall contain the following controls:

3.5.1.5.1 Present Position Latitude and Longitude - Present position latitude and longitude controls shall be provided to set the present position latitude and longitude counters at the desired values. These controls shall be a push-to-turn manual control.

3.5.1.5.2 Destination Latitude and Longitude - Controls shall be provided to adjust the destination latitude and longitude counters to the desired values. These controls shall be a combination push-to-turn manual control and two speed electrical slew. The manual push-to-turn controls shall be useable when the master selector switch is in the OFF and STBY positions. The two speed electrical slew controls shall be useable when the master selector switch is in the TEST, D1, and D2 positions. In addition, one speed electrical slew control shall be available when the master selector switch is in the STBY mode.

3.5.1.5.2.1 Electrical Slew Controls - The destination latitude and longitude electrical slew controls shall be capable of changing the destination latitude counter north and south and of changing the destination longitude counter east and west. The electrical slew shall have two speeds. The fast speed electrical slew shall be 45°/min ± 5°/min for D1 and D2. The slow speed electrical slew shall be 10°/min ± 2°/min for D1 and D2. The slew speed shall be 40°/min ± 15°/min for STBY.

3.5.1.5.3 Magnetic Variation - A control shall be provided to adjust the magnetic variation counter to the desired value. The control shall be the manual push-to-turn type.

3.5.1.5.4 Wind Speed and Direction - Controls shall be provided to adjust the wind speed and the wind direction counters. These controls shall be the manual push-to-turn type.

3.5.1.5.5 Master Selector Switch - The master selector switch shall control the operation of the Navigation Computer Set and shall have five positions: TEST, OFF, STBY, D1, D2.

(1) TEST: A standard problem shall be applied to verify proper operation of the computer set.

(2) OFF: All electric power shall be removed from the equipment.

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(3) STBY: Electric power shall be applied to all parts of the computer set except the integrators.

(4) D1: Electric power shall be applied to the equipment and computation based on destination coordinates D1 displayed on the destination position counters.

(5) D2: Electric power shall be applied to the equipment and computation based on destination coordinates D2 displayed on the destination position counters.

3.5.1.6 Electrical Connections - Electrical Connections shall be provided as shown in Figure 1.

3.5.1.7 Contents of Unit - The Control-Indicator shall contain the following sub-assemblies:

(1) Magnetic Variation Module

(2) Wind Speed and Direction Module

(3) Longitude Module

(4) Latitude Module

3.5.2 Computer, Navigational CP-774/ASN-41 - The Computer, Navigational, shall meet the following requirements:

3.5.2.1 Function - The navigation computer shall include all of the computer set's power supply and all electro-mechanical computer modules not associated with front panel controls and/or displays. Thus this unit shall provide computation tasks required with inputs from other aircraft systems as well as the control-indicator.

*3.5.2.2 Form Factor - the Navigational Computer shall be enclosed in a case and shall conform to the dimensions shown in Figure 3.

3.5.2.3 Weight - The weight of the Navigational Computer Unit shall not exceed 23 pounds.

3.5.2.4 Controls and Displays - The Navigational Computer Unit shall not contain any controls or displays.

3.5.2.5 Electrical Connections - Electrical connections shall be provided as shown in Figure 1.

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3.5.2.6 Contents of Unit - The computer unit shall contain the following sub-assemblies:

(1) True Heading and Drift Angle Module

(2) Ground Speed and True Air Speed Module

(3) Integrator Module

(4) Course Angle and Distance Module

(5) Memory Module

(6) Power Supply Module

3.5.3 Mounting MT3491/ASN-41 - The mounting shall meet the following requirements:

3.5.3.1 Function - The mounting shall provide shock and vibration isolation and mounting provisions for the Navigational Computer Unit. The mounting shall meet the applicable requirements of Specification MIL-C-172.

*3.5.3.2 Form Factor - The maximum allowable dimensions of the mounting when used with the navigational computer shall be as shown in Fig. 4.

3.5.3.3 Weight - The weight of the mounting shall not exceed 2½ pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

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4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

* 4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 24 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (encluding resonant points) and a minimum amplitude of ±2 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 24-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used.

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During the (24) hour period of operation following the ten minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The (24) hour period specified above may be composed of periods of four or more hours to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production on Reliability Tests". For the Sampling Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot.

4.4.3.3 Length of Heat Portion of Cycle - During each cycyle the equipment shall be operated during the heating portion of the cycle. After stabilization at the high temperature limit required by the test level, the equipment shall be operated for a period of 6 hours.

* 4.4.3.4 Performance Characteristics to be Measured -

Characteristic	Specification Requirement	Time in Cycle that Measurement is to be Made
Present Position Latitude and Longitude	1 NM per hour or P (Distance traveled) whichever is greater where P=±0.25% in latitude computation P=±0.25% for longitude computation from 60° N to 60° S P=±0.50% for longitude computation from 60° to 70° N & S P=±1% for longitude computation from 70° to 80° N & S	Once each heat cycle

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Wind Speed	+1.0 knot or 0.3% of ground speed
Wind Direction	±1° or compatible with wind speed tolerance
Distance to Destination	0 to 195 miles: 1.5 NM or 1.5% of distance whichever is greater
	195 to 1999 miles: 7.5 NM + 1.5% of distance
Relative Bearing to Destination	±1.5° or consistent with distance tolerance

4.4.3.5 Failure Criteria - In addition to the requirements of MIL-STD-781, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests no preventive maintenance measures may be performed upon the equipment.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 + 5 volts (at lowest applicable frequency)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

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(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The Navigation Computer Set is intended for use in Naval fixed wing aircraft. The Navigation Computer Set is to solve the navigation problem, that is, compute present position and distance and course to a selected destination. The Navigation Computer Set shall be used as part of a navigation system consisting of doppler type ground velocity sensor, a heading reference, a true air speed computer and a bearing-distance-heading indicator.

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6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-Repairable Sub-Assemblies - As a general rule, non-repairable sub-assemblies should be encapsulated or hermetically-sealed. The number of connections internal to the sub-assembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the sub-assembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable sub-assembly should evidence a Mean-Time-to-Failure greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications reference therein. All referenced specifications shall apply to the extent specified.

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6.6 Type Designation - The parenthesis (*), when used in the type designation of equipment, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing.(see 5.1).

6.8 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

* 6.9 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Control-Indicator	C-6243(*)ASN-41	3.5.1
Computer, Navigational	CP-774(*)ASN-41	3.5.2
Mounting	MT-3491/ASN-41	3.5.3

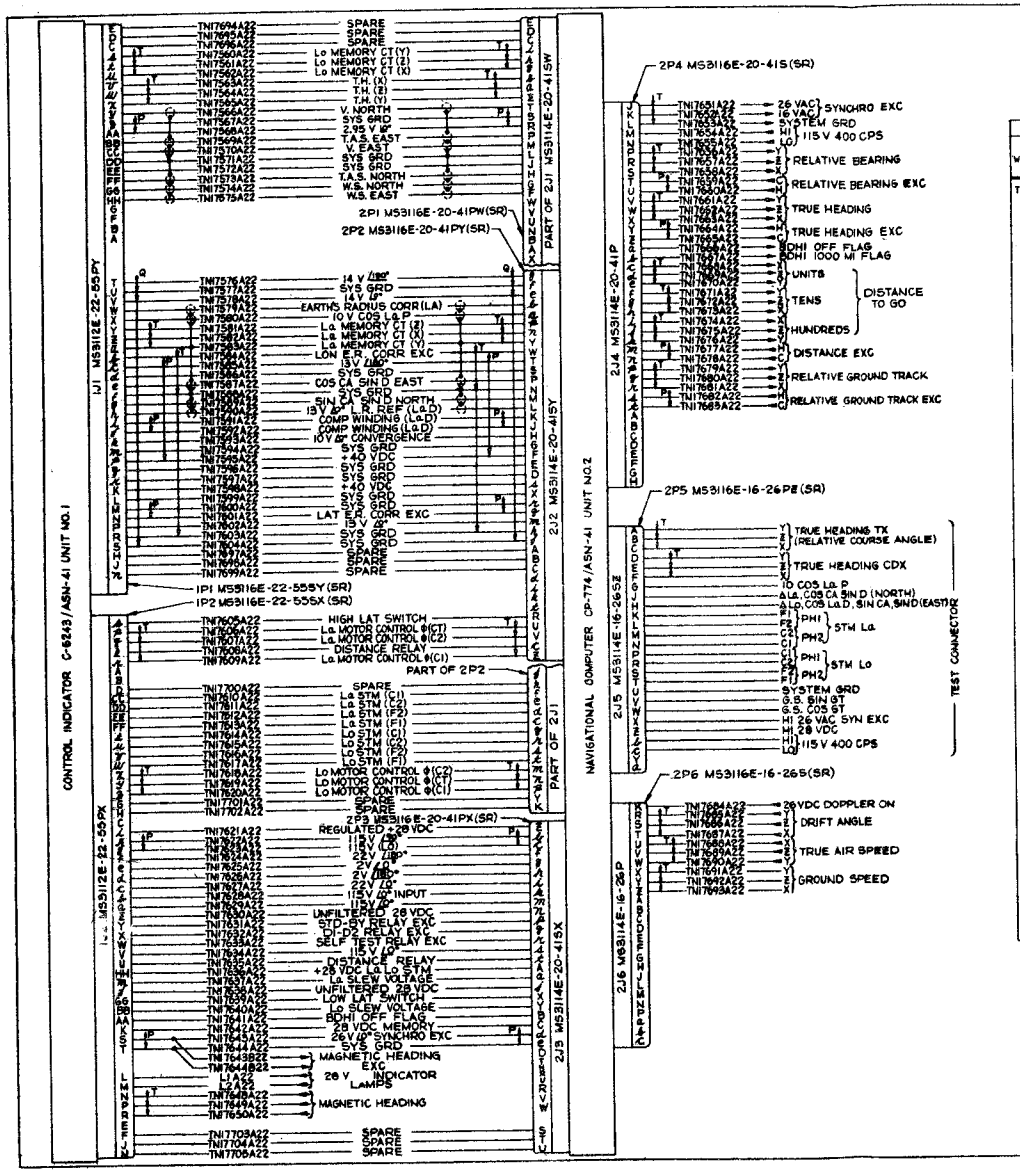
6.9.1 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Navigation Set	AN/APN-153(V)
Heading Reference Computer Set	AN/AJB-3(*)
Indicator	ID-663/U
Heading Reference System	AN/ASN-50
Compass	MA1
Compass	G2
Central Air Data Computer System	
Central Gyro Reference System	

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WIRE TABLE

WIRE NO.	SYSTEM OPERATING VOLTAGE	CONDUCTED CURRENT AMPERES	WIRE NO.	SYSTEM OPERATING VOLTAGE	CONDUCTED CURRENT AMPERES
TN17560	11.8	.070	TN17634	115	1.5
561	11.8	.070	635	0	.1
562	11.8	.070	636	28 VDC	.16
563	11.8	.3	637	22	.15
564	11.8	.3	638	28 VDC	.5
565	11.8	.3	639	0	.05
566	10	.001	640	22	.15
567	0	.007	641	0	NOTE 8
568	2.95	.007	642	28	.15
569	9.8	.001	643	26	.22
570	10	.001	6438	26	NOTE 8
571	0	.001	644	0	.22
572	0	.001	644B	0	NOTE 8
573	9.8	.001	648	11.8	.3
574	2.95	.001	649	11.8	.3
575	2.95	.001	650	11.8	.3
576	14	.006	651	26	NOTE 8
577	0	.006	652	16	NOTE 8
578	14	.006	653	0	0
579	.1	.001	654	115	1.5
580	10	.001	655	0	.5
581	11.8	.070	656	11.8	NOTE 8
582	11.8	.070	657	11.8	NOTE 8
583	11.8	.070	658	11.8	NOTE 8
584	.04	.001	659	0	NOTE 8
585	13	.01	660	26	NOTE 8
586	0	.01	661	11.8	NOTE 8
587	13	.001	662	11.8	NOTE 8
588	0	.001	663	11.8	NOTE 8
589	13	.001	664	26	NOTE 8
590	13	.001	665	0	NOTE 8
591	40 VDC	.010	666	0	NOTE 8
592	40 VDC	.010	667	0	NOTE 8
593	10	.001	668	11.8	NOTE 8
594	0	.001	669	11.8	NOTE 8
595	440 VDC	.05	670	11.8	NOTE 8
596	0	.05	671	11.8	NOTE 8
597	0	.05	672	11.8	NOTE 8
598	40 VDC	.05	673	11.8	NOTE 8
599	0	.001	674	11.8	NOTE 8
600	0	.001	675	11.8	NOTE 8
601	.1	.001	676	11.8	NOTE 8
602	13	.001	677	26	NOTE 8
603	0	.001	678	0	NOTE 8
604	0	.006	679	11.8	NOTE 8
605	28 VDC	.050	680	11.8	NOTE 8
606	28 VDC	.15	681	11.8	NOTE 8
607	20	.145	682	26	NOTE 8
608	0	.1	683	0	NOTE 8
609	20	.145	684	26 VDC	.05
610	28 VDC	.04	685	11.8	.035
611	28 VDC	.04	686	11.8	.035
612	28 VDC	.04	687	11.8	.035
613	28 VDC	.04	688	11.8	.035
614	28 VDC	.04	689	11.8	.035
615	28 VDC	.04	690	11.8	.035
616	28 VDC	.04	691	11.8	.035
617	28 VDC	.04	692	11.8	.035
618	20	.145	693	11.8	.035
619	28 VDC	.85	694	0	0
620	20	.145	695	0	0
621	28 VDC	.020	696	0	0
622	115	.2	697	0	0
623	0	.2	698	0	0
624	22	.15	699	0	0
625	2	.015	700	0	0
626	2	.015	701	0	0
627	22	.15	702	0	0
628	115	1.5	703	0	0
629	115	.1	704	0	0
630	28 VDC	.5	705	0	0
631	28 VDC	.15	L1	28	.25
632	28 VDC	.15	L2	0	.25
633	28 VDC	.15			

NOTES

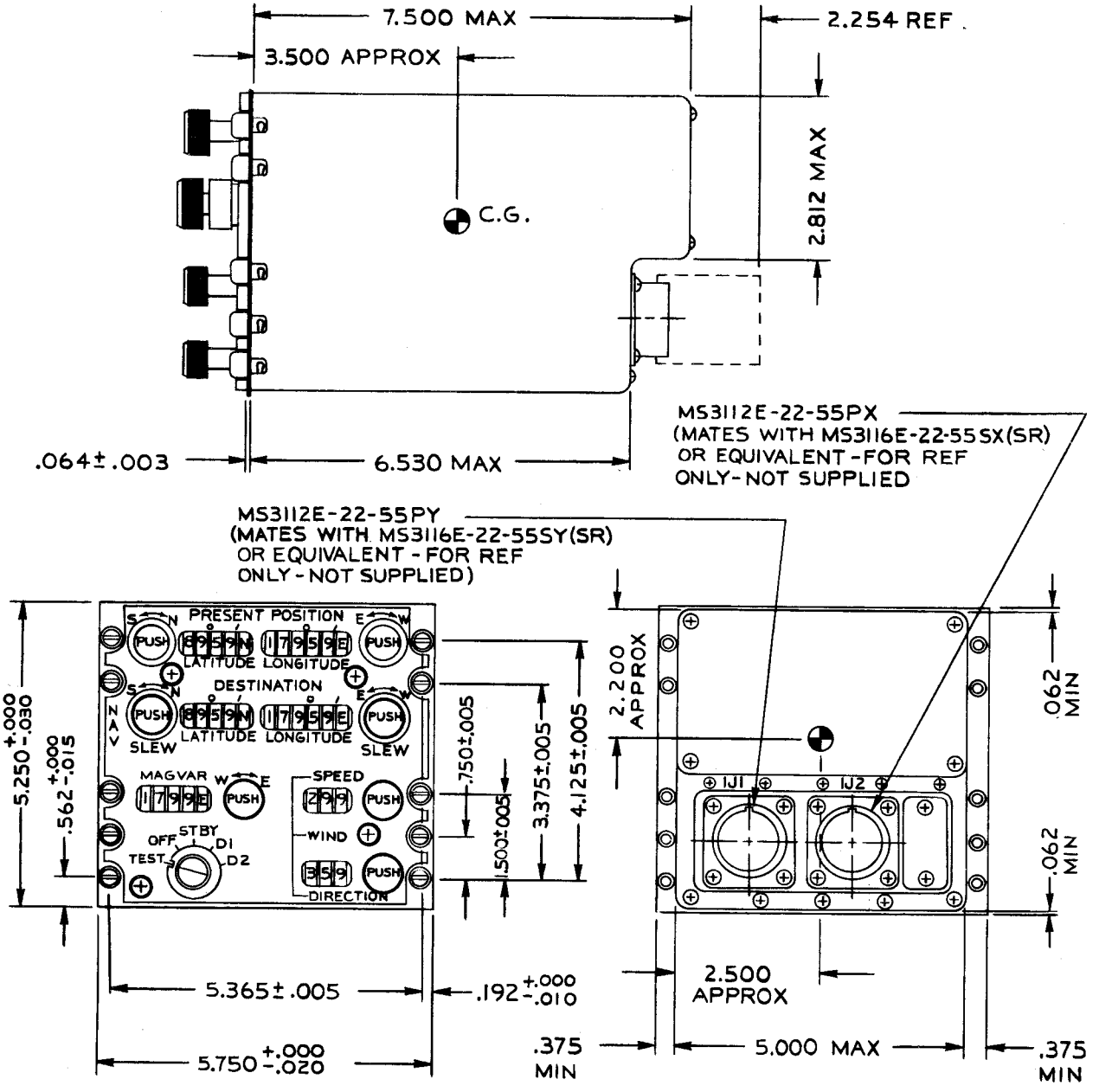
1. INSTALLATION OF WIRING SHALL CONFORM TO SPECIFICATION MIL-W-5088.
2. THE BLOCK OF WIRE CODE NUMBERS ASSIGNED TO THE AN/ASN-41 IS TN17560 THRU TN17784. THE HIGHEST NUMBER USED IS TN17705.
3. SEGMENT LETTERS IN ADDITION TO THOSE SHOWN MAY BE USED AS NECESSARY TO FIT THE PARTICULAR INSTALLATION.
4. WIRE CODING FOR PANEL LIGHTING WIRES SHALL BE ASSIGNED BY THE INSTALLING ACTIVITY IN NEW AIRCRAFT THIS WILL BE AS SEGMENTS OF THE PANEL LIGHTING CIRCUIT USING LETTER "L" PER SPECIFICATION MIL-W-5088. FOR OLD AIRCRAFT THIS WILL BE AS SEGMENTS OF THE EXISTING CODING ON WIRES AT THE POINT OF TIE-IN .
5. WIRE SIZES SHOWN ARE MINIMUM SIZES. LARGER WIRES SHALL BE USED WHEN REQUIRED.
6. ALL CONNECTOR PINS ARE SHOWN WHETHER USED OR NOT, AND THE REQUIRED SPARES WITH WIRE CODING AND WIRE SIZES.
7. PLUG IDENTIFICATION IS REQUIRED ONLY IN THE CASE OF SPECIAL PLUGS.
8. CONDUCTED CURRENT DEPENDENT UPON USING EQUIPMENT.
9. THE LETTER "P" DENOTES A TWISTED PAIR, THE LETTER "T" A TRIPLE TWIST AND THE LETTER "Q" A QUADRUPLE TWIST.
10. SHIELDED WIRES SHALL CONFORM TO SPECIFICATION MIL-W-1678.

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INTERCONNECTION DIAGRAM (AN/ASN-41)	DWG NO. FIGURE 1	REV

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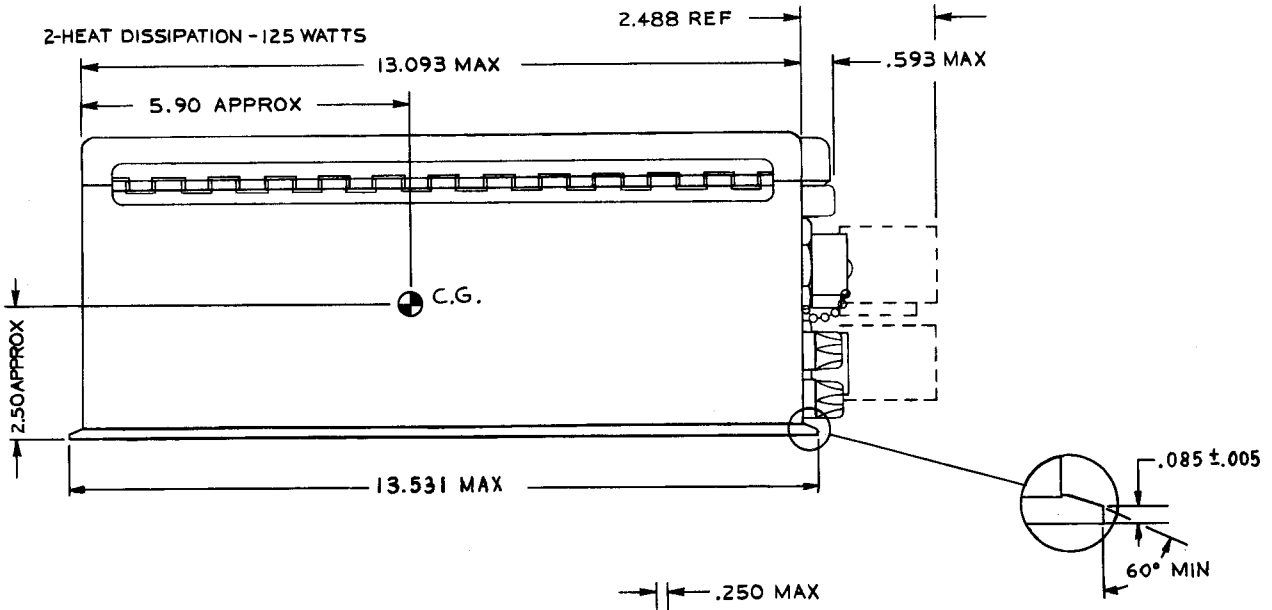
NOTES:
UNLESS OTHERWISE SPECIFIED
1. UNIT WEIGHT - 10.0 LBS
2. HEAT DISSIPATION - 75 WATTS

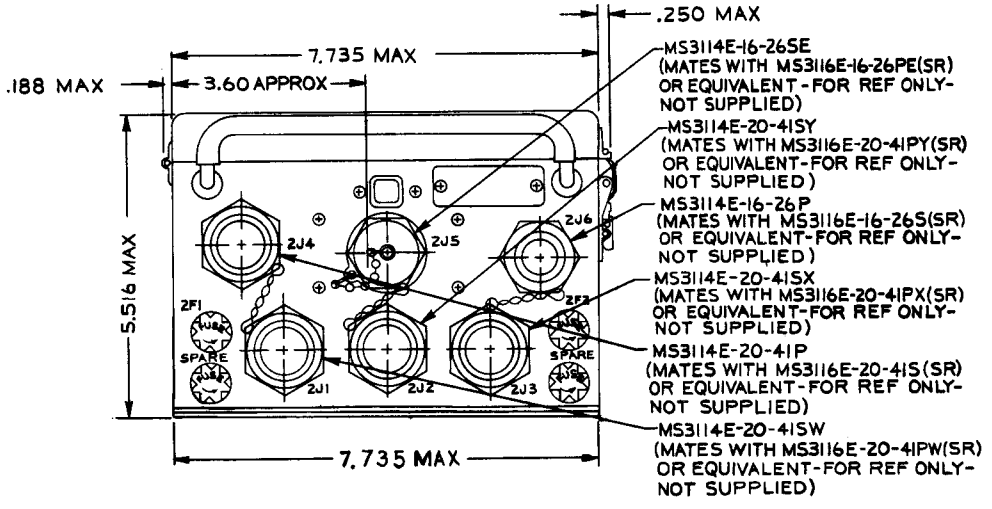


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CONTROL INDICATOR OUTLINE DIMENSIONS	DWG NO. FIGURE 2	REV

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NOTES:
UNLESS OTHERWISE SPECIFIED
1-UNIT WEIGHT - 23 LBS
2-HEAT DISSIPATION - 125 WATTS
3-VENTILATION SPACE - NONE REQD



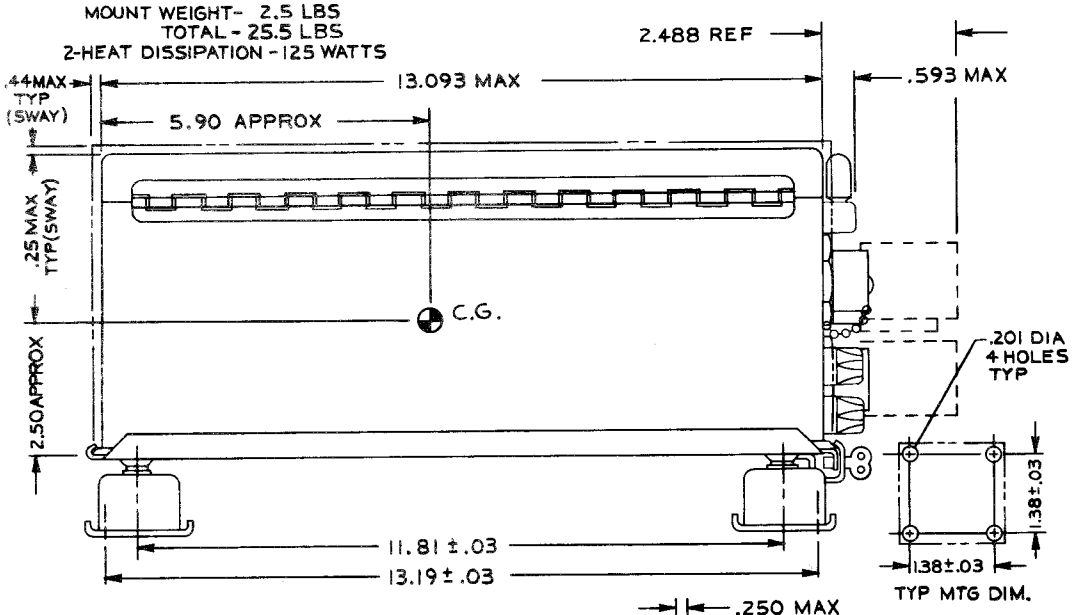


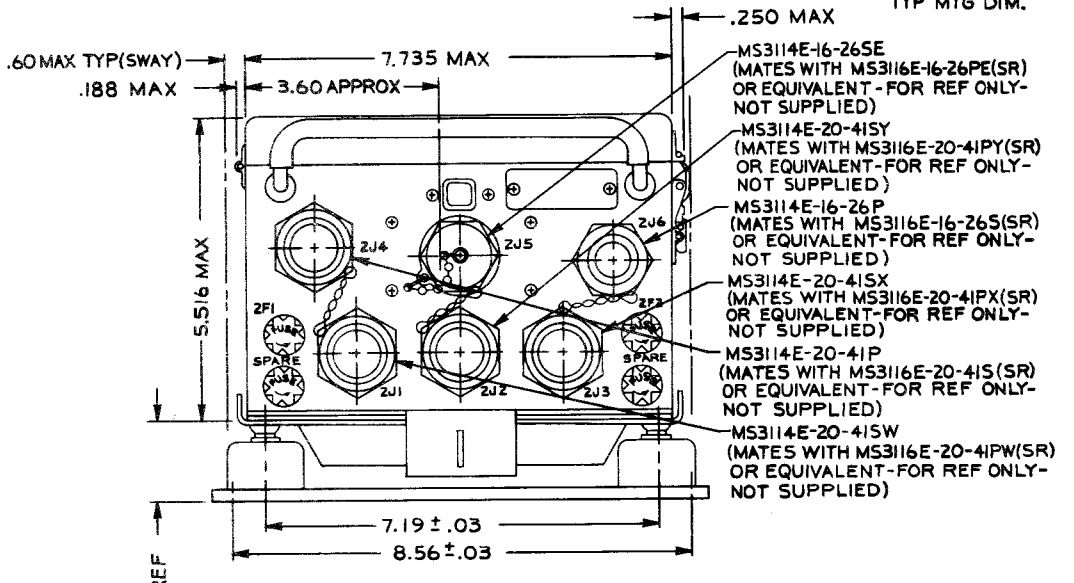
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COMPUTER, NAVIGATIONAL OUTLINE DIMENSIONS	DWG NO. FIGURE 3	REV

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NOTES:
UNLESS OTHERWISE SPECIFIED
1-UNIT WEIGHT - 23 LBS
MOUNT WEIGHT- 2.5 LBS
TOTAL - 25.5 LBS
2-HEAT DISSIPATION - 125 WATTS
3-VENTILATION SPACE - NONE REQD
4-SWAY SPACE DIMENSIONS ARE SHOWN





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COMPUTER, NAVIGATIONAL OUTLINE DIMENSIONS	DWG NO. FIGURE 4	REV

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-81185B(AS) COMPUTER SET, NAVIGATIONAL AN/ASN-41(*)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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